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Original Article

Ablation of CD22 induces a pro-inflammatory transcriptome shift and endocytic deficits in M2-like macrophages

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Abstract

The interactions between environmental glycans and glycan-binding proteins modulate a host of processes across biological systems. The Siglec/sialic acid axis has gained increasing attention as an immunologic checkpoint due to its involvement with reducing inflammatory processes and promoting tumor growth. Siglec-2, or CD22, has been extensively characterized as a co-receptor for the B cell receptor (BCR) and is critical for the prevention of self-reactive B cell responses through its recognition of α2,6-linked sialic acids. More recently, CD22 has emerged as an important receptor for macrophage biology. Here, we investigate the consequences of genetic ablation of CD22 in murine macrophages (CD22KO). Aged CD22KO mice developed a fatty liver phenotype similar to that seen in aged animals lacking hepatocyte α2,6-sialylation (HcKO). CD22KO bone marrow-derived macrophages (BMDMs) exhibited few differences in canonical markers of M1-like and M2-like polarization, but M2-like CD22KO BMDMs showed a pro-inflammatory shift in transcriptome and a reduction in endocytic and efferocytotic capacity. These data suggest that CD22 in murine M2-like macrophages is strongly associated with a homeostatic transcriptional profile and directly participates in immunologically silent housekeeping functions such as clearance of sialylated-self debris through the Siglec-sialic acid axis.

Keywords: CD22; endocytosis; macrophage; sialic acid; siglec.

Introduction

Sialic acids are implicated in an array of biological systems and processes, often acting as a "switch" to modulate cellular responses (Varki and Gagneux 2012). The nature of this modulation can be impacted by many factors, including the linkage of sialic acid to the galactose, N-acetyl-galactosamine, or sialic acid on the underlying glycan (i.e. α2,3, α2,6, or α2,8) and modifications such as sulfation, glycolylation, and acetylation (e.g. Neu5Gc versus Neu5Ac) (Ereño-Orbea et al. 2017; Lübbers et al. 2018; Gonzalez-Gil and Schnaar 2021). The addition of sialic acid in an α2,6 linkage to an underlying galactose residue is mediated by the enzyme ST6Gal1 (Paulson et al. 1977; Martin et al. 2002), and both α2,6-linked sialic acids and ST6Gal1 are generally recognized as anti-inflammatory switches in many immunological processes (Jones 2018; Zhou et al. 2018).

Siglecs (sialic acid-binding immunoglobulin-like lectins) are a family of receptors expressed on a variety of cells of the hematopoietic system (Crocker et al. 2007). These receptors mediate their functions through engaging sialic acids in their environment via an extracellular carbohydrate recognition domain (CRD), typically resulting in signal transduction and downstream cellular processes (Crocker et al. 2007; Zhou et al. 2018). Siglecs have varying specificities for different sialic acids and their glycosidic linkages depending on the conformation of the CRD. The majority of Siglecs contain immunoreceptor tyrosine-based inhibitory motifs (ITIMs) within their intracellular signaling tails. Upon CRD ligation to a sialylated ligand, these ITIMs become phosphorylated and recruit phosphatases, generally resulting in the inhibition of kinase-dependent signaling cascades. Thus, many Siglecs are considered to be inhibitory.

Further underscoring their immunomodulatory properties, Siglecs are increasingly implicated as immunologic checkpoints within tumor microenvironments (TMEs) for their promotion of immunosuppressive environments (Stanczak and Läubli 2023). Similarly, ST6Gal1 is considered an oncogene and is upregulated by many tumors to promote stemness among cancer stem cells (Schultz et al. 2016), promote metastasis (Britain et al. 2021), and drive an immunosuppressive environment through the Siglec/sialic acid axis to promote the dampening of immune cascades, thereby evading anti-tumor responses of the immune system (Beatson et al. 2016). As a result, Siglecs are becoming attractive targets for immunotherapies aimed at blocking the Siglec/sialic acid interaction to shift the TME to a more pro-inflammatory/anti-tumor state (Duan and Paulson 2020; Lim et al. 2021; Läubli et al. 2022).

Building upon the evidence that sialylation within the local micro-environment drives immunosuppression, our group previously demonstrated that a mouse lacking α2,6-sialylation in hepatocytes through conditional ablation of ST6Gal1 (HcKO) is skewed towards a pro-inflammatory

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hematopoietic compartment. Moreover, we discovered that the Siglec/sialic acid axis is critical for mediating this effect (Oswald et al. 2020a; Oswald et al. 2020b). In the absence of inhibitory ligation provided by sialic acid ligands, a pro-inflammatory shift in tissue-resident macrophages and systemic cells such as splenic T cells results. These changes in the hematopoietic compartment drive a non-alcoholic fatty liver phenotype in aged HcKO animals, suggesting that the pro-inflammatory skew of HcKO liver macrophages (Kupffer cells) results in overt liver pathology.

Siglec-2 (CD22) is unique among the Siglecs for its strong preference for the α2,6 linkage of sialic acids (Crocker et al. 2007). CD22 has been extensively characterized for its critical roles in B cell biology (Clark and Giltiay 2018). CD22 is a co-receptor for the B cell receptor (BCR), essentially serving as a "brake" to prevent self-reactive B cell responses (Walker and Smith 2008). Loss of CD22 results in hyper-reactive B cell populations and increased concentration of autoantibodies, underscoring the notion that CD22 is integral for proper B cell function (O'Keefe et al. 1996). However, more recently, CD22 has emerged as an important receptor in macrophage biology (Pluvinage et al. 2019; Aires et al. 2021; Kukan et al. 2024).

Macrophages are phagocytic cells of the innate immune system that exist on a wide and diverse spectrum from pro-inflammatory M1-like macrophages, driven by IFNγ and LPS stimulation, to anti-inflammatory M2-like macrophages, primarily driven by IL-4 stimulation (Strizova et al. 2023). Macrophage polarization state influences the phenotype and function of these cells, with M1-like macrophages characterized by promoting inflammatory cascades, driving T cell activation through antigen presentation, and participating in phagocytosis and microbial killing, while M2-like macrophages tend to dampen inflammation and mediate wound healing and other pro-resolving processes. Our group has discovered that CD22 is strongly associated with the M2-like, anti-inflammatory phenotype of macrophages. For example, after 48 h of IL-4-mediated polarization, robust CD22 expression is induced among bone marrow-derived macrophages (BMDMs). More importantly, we found that CD22 assists in the immunologically-silent clearance of sialylated debris (endocytosis) and apoptotic cells (efferocytosis), preventing the exacerbation of inflammation (Kukan et al. 2025).

While many studies have documented the impact of CD22 knockout in B cells, little is known about the impact of CD22 ablation in macrophages. Our previous work compared BMDMs expressing ($CD22^+$) or lacking ($CD22^-$) CD22 on a wild-type (WT) background; however, the effects of genetic knockout of CD22 (CD22KO) in macrophages have remained unexplored. In the present study, we describe the effects of CD22 knockout on macrophage phenotype and function with a specific focus on the 48 h-polarized M2-like BMDM group due to their demonstrated high expression of CD22. We found that loss of CD22 has broad pro-inflammatory implications for macrophage biology, and like their WT M2-like $CD22^-$ macrophage counterparts (Kukan et al. 2025), M2-like CD22KO macrophages showed reduced endocytosis and efferocytosis of sialylated glycoproteins and apoptotic cells, respectively. These findings highlight the importance of CD22 on macrophages in sensing the local tissue environment to maintain optimal homeostatic immunologic tone.

Results

HcKO leukocytes are more pro-inflammatory

Our previous studies of unchallenged HcKO mice showed increased percentages of antigen-experienced ($CD44^+$) T cells and development of an age-dependent spontaneous fatty liver phenotype (Oswald et al. 2020a; Oswald et al. 2020b). To better understand the mechanisms underlying these changes, we began by assessing the recall response of young and aged WT and HcKO $CD4^+$ T cells. In T cells isolated from the livers of young mice, no differences in T cell activation ability were observed between genotypes (Fig. 1A). Conversely, HcKO T cells from aged mice showed increased recall activation as compared to WT T cells from aged mice (Fig. 1B). Complementing this observation, T cells isolated from aged HcKO spleens showed a decrease in naïve (Tn; $CD44^{low}CD62L^{high}$) $CD4^+$ T cells and a concomitant increase in effector memory (Tem; $CD44^{high}CD62L^{low}$) $CD4^+$ T cells (Fig. 1C). Importantly, T cells isolated from aged CD22 knockout (CD22KO) spleens showed a similar decrease in Tn cells and a concomitant increase in Tem cells, suggesting that both the lack of CD22 (CD22KO) and the lack of CD22 ligand in the form of α2,6-linked sialic acids (HcKO) result in increased antigen-experienced T cells (Fig. 1D). These data correlate with our published findings in which T cells from HcKO animals older than six months of age show greater antigen experience compared to WT controls by demonstrating an increased pro-inflammatory recall response upon T cell receptor activation, as determined by activation sensitivity and expression of activation markers.

Combined with our prior publications (Oswald et al. 2020a; Oswald et al. 2020b), these data suggest that antigen presentation leading to T cell activation by liver cells must be altered in the absence of α2,6-sialylated glycans on hepatocytes. In fact, initial observations in the HcKO model revealed a pro-inflammatory skew among Kupffer cells in aged mice (Oswald et al. 2020a). In order to determine whether changes in the liver macrophages themselves, resulting from the loss of α2,6-sialylated glycans in the localized microenvironment, were sufficient to differentially drive T cell activation, we isolated macrophages from the livers of aged WT and HcKO animals, cultured them with OT-II T cells with or without Ova protein, and measured cytokine production in the supernatant. As predicted, HcKO macrophages elicited greater pro-inflammatory cytokine production from T cells than did WT macrophages, including IFNγ, LIX, MIP-1α, and MIP-1β (Fig. 1E). This increased tendency to present antigen and activate T cells by liver macrophages provides a plausible explanation for the in vivo increases in T cell antigen experience in old HcKO animals (Oswald et al. 2020b) and suggests that this difference is driven by sensing the absence of α2,6-linked sialic acid ligands in the liver microenvironment of these mice.

Since CD22 is a known and uniquely specific receptor for α2,6-linked sialic acid ligands, we compared the aged phenotype of WT, HcKO, and CD22KO mice. This was done to directly interrogate the physiologic consequences of removing the ligand (HcKO) versus a known receptor (CD22KO). As a control, the glycosylation of circulating plasma glycoproteins was analyzed and found to remain largely the same in aged CD22KO as in aged WT animals (Figs S1 and S2), unlike in the HcKO mouse (Oswald et al. 2020a). Interestingly, aged CD22KO animals phenocopied the fatty liver phenotype seen

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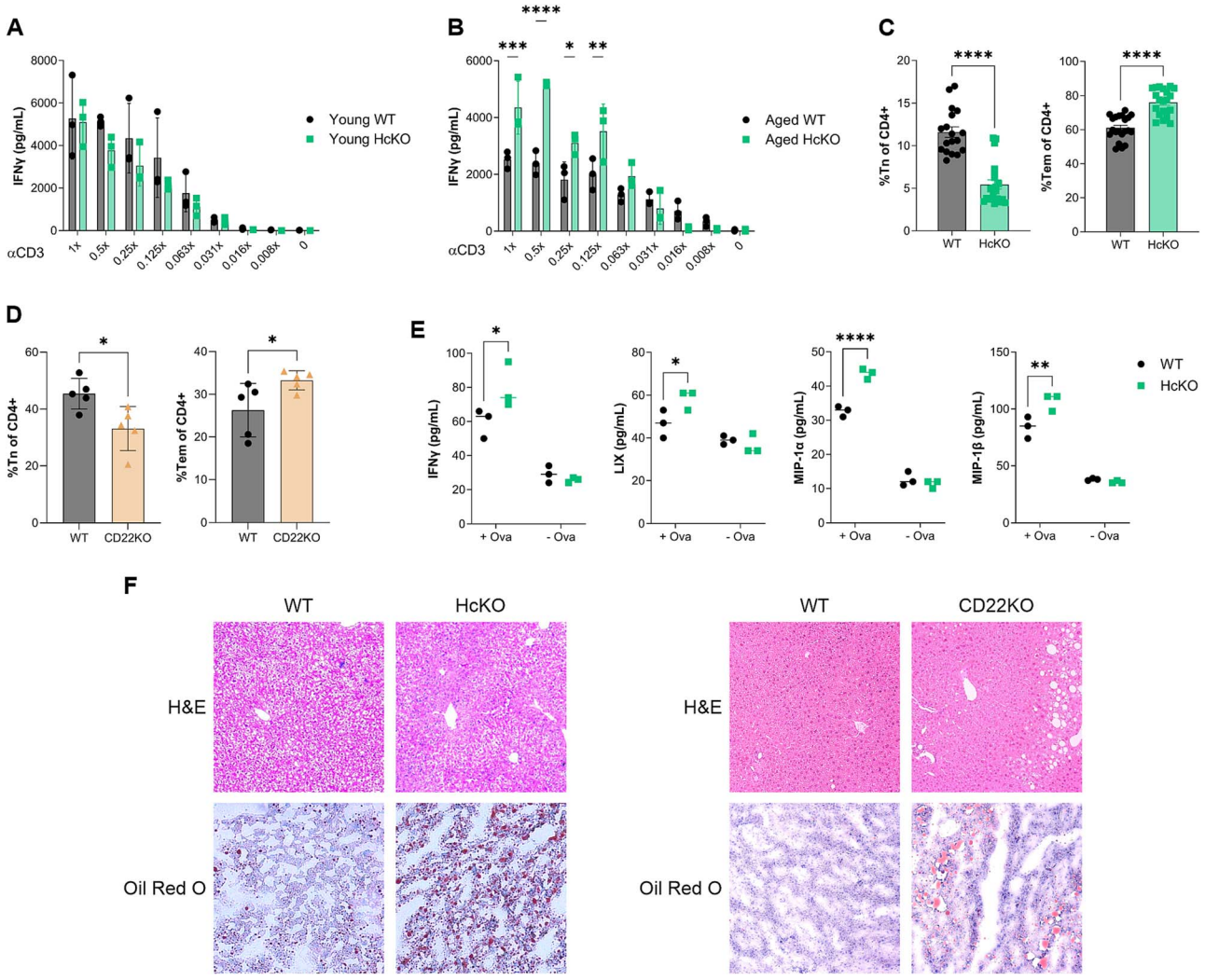


Fig. 1. HcKO leukocytes are more pro-inflammatory. IFNγ ELISA of T cells stimulated with various concentrations of αCD3 isolated from young A) or aged B) WT or HcKO mice ($n = 3$). C) Flow cytometry data showing Tn ($CD44^{low}CD62L^{high}$) and tem ($CD44^{high}CD62L^{low}$) isolated from aged WT and HcKO mouse spleens ($n \geq 18$). D) Flow cytometry data showing Tn and tem isolated from aged WT and CD22KO mouse spleens ($n = 5$). E) Cytokine production measured via Luminex from T cells co-cultured with aged WT or HcKO macrophages in the presence or absence of antigen (ova) ($n = 3$). F) Representative images showing liver histology via H&E and oil red O staining of aged WT versus aged HcKO (left) and aged WT versus aged CD22KO (right) ($n = 3$). Statistical analyses are two-way ANOVA A, B, E) or Student's t-test (C, D). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. All other comparisons not significant ($P \geq 0.05$).

in aged HcKO animals. Specifically, aged CD22KO animals showed similarly increased liver fat deposits, as demonstrated by H&E and Oil Red O staining (Fig. 1F), relative to the WT livers.

Collectively, these data support a model in which α2,6-sialylation upon hepatocytes is sensed by CD22 on macrophages and that loss of these glycans (HcKO) or their specific receptor (CD22) leads to fatty liver disease through inappropriate immune activation over time.

Loss of CD22 does not impact BMDM polarization

Because aged CD22KO animals developed a fatty liver phenotype, similar to aged HcKO animals, we hypothesized that loss of CD22 may drive a pro-inflammatory skew in macrophages. We first measured protein and transcript expression by flow cytometry and qPCR, respectively, of canonical markers of M1-like and M2-like macrophage phenotype. Given our prior observations that 48 h of IL-4 stimulation (M2-like polarization) drives robust CD22 expression in BMDMs, we measured expression of these polarization markers at 48 h of polarization in both WT and CD22KO macrophages. Flow cytometry analysis revealed similar expression of CD86 in WT and CD22KO M1-like BMDMs, with CD22KO BMDMs showing only a minor decrease in CD86 expression by MFI and no difference by percent positivity (Fig. 2A). Consistent with this observation, 48 h-polarized M2-like CD22KO BMDMs also showed no difference in expression of CD206 and CD301, two canonical markers of M2-like polarization, suggesting that BMDM polarization is generally unaffected by CD22 ablation (Fig. 2B). We further confirmed this using qPCR to measure transcript expression of a variety of M1-like and M2-like polarization markers. CD22KO BMDMs polarized to M1-like for 48 h showed slightly decreased iNOS transcript expression but no difference in CD86 transcript expression (Fig. 2C). When polarized to M2-like for 48 h, CD22KO BMDMs again showed no difference in transcript expression of common M2-like markers CD206, CD301,

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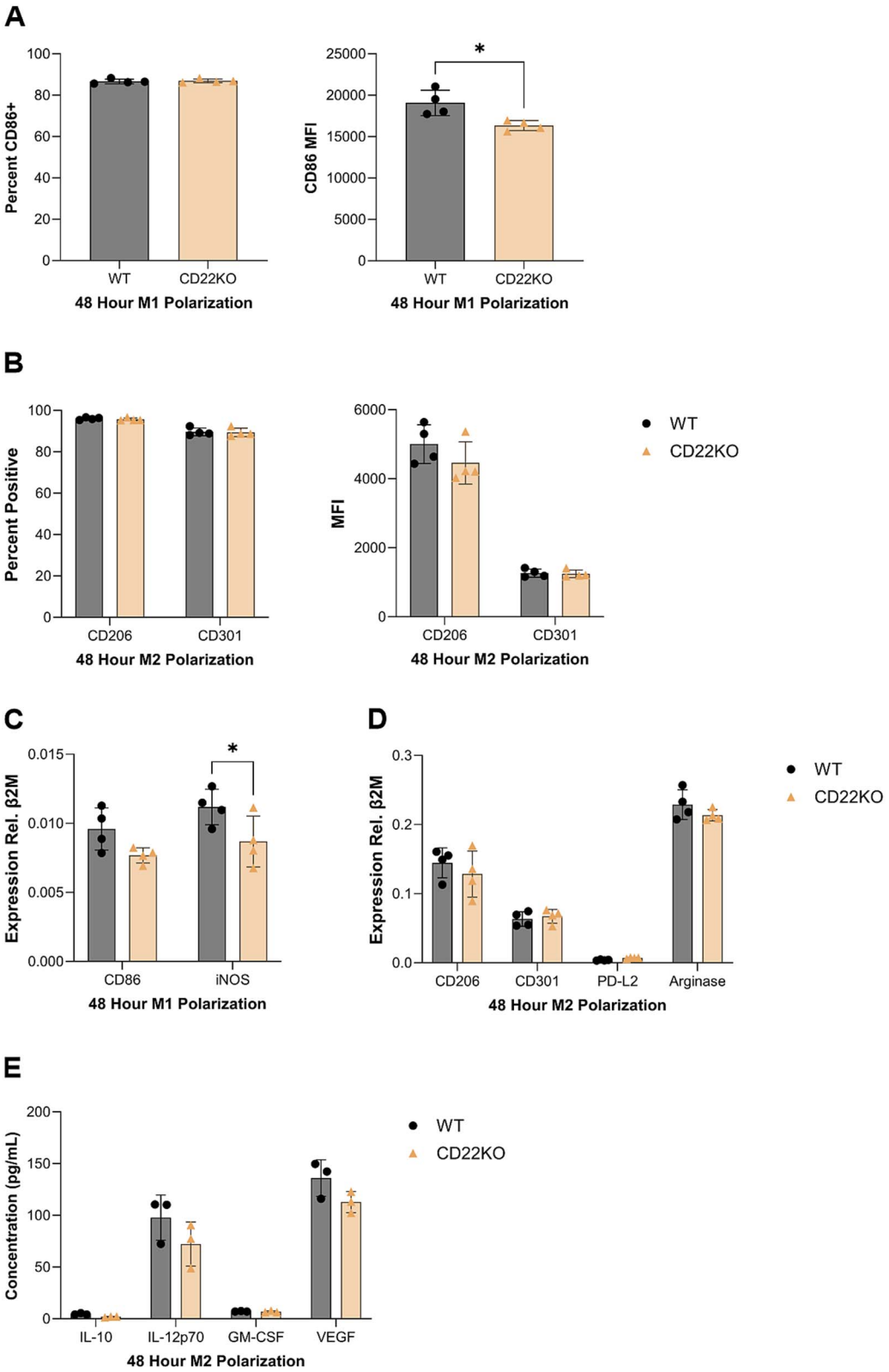


Fig. 2. Loss of CD22 does not impact BMDM polarization. A) Flow cytometry data showing percent positive (left) and MFI (right) of CD86 for F4/80$^+$ WT and CD22KO macrophages polarized to an M1-like phenotype for 48 h ($n = 4$). B) Flow cytometry data showing percent positive (left) and MFI (right) of CD206 and CD301 of WT and CD22KO in F4/80$^+$ BMDMs polarized to an M2-like phenotype for 48 h ($n = 4$). C) qPCR data showing transcript expression of CD86 and iNOS in WT and CD22KO BMDMs polarized to M1-like for 48 h ($n = 4$). D) qPCR data showing transcript expression of CD206, CD301, PD-L2, and arginase in WT and CD22KO BMDMs polarized to M2-like for 48 h ($n = 4$). E) Luminex data showing cytokine production in WT and CD22KO 48 h-polarized M2-like BMDMs ($n = 3$). Statistical analyses are Student's t-test A) or two-way ANOVA B–E). *$P < 0.05$. All other comparisons not significant ($P \geq 0.05$).

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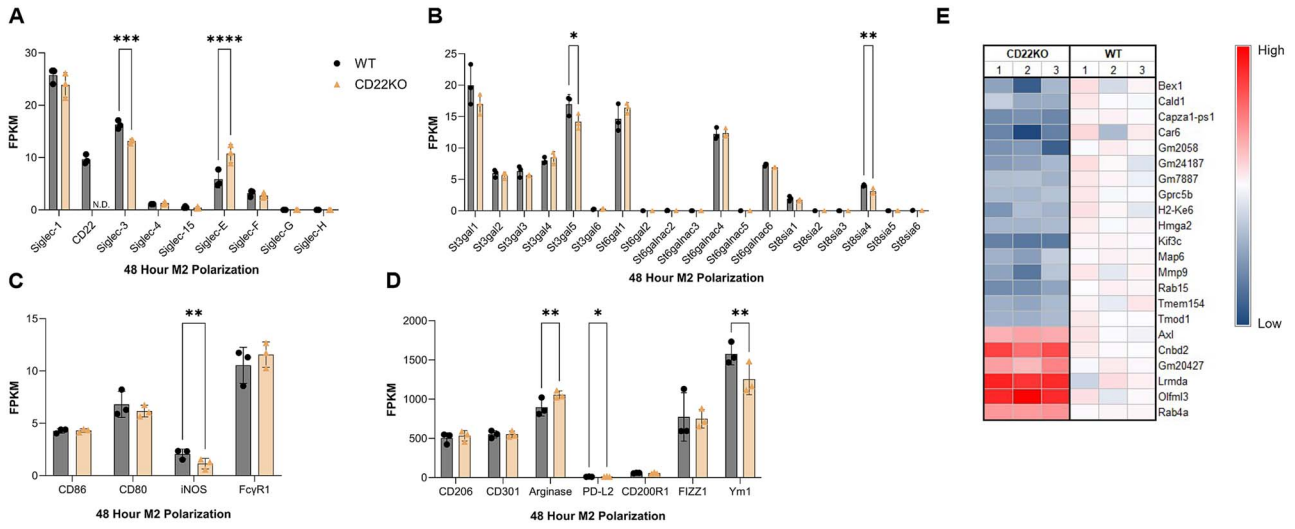


Fig. 3. CD22KO M2-like BMDMs exhibit transcriptional changes. Fragments per kilobase per million mapped fragments (FPKM) values for Siglecs A), sialyltransferases B), M1-like polarization markers C), and M2-like polarization markers D) in WT and CD22KO BMDMs polarized to M2-like for 48 h ($n = 3$). E) Heatmap from RNAseq analysis displaying differentially expressed genes in WT and CD22KO BMDMs polarized to M2-like for 48 h ($n = 3$). Two genes (Hist1h2al and Nnt) were excluded from the heatmap despite meeting our threshold for inclusion because Hist1h2al showed undetectable expression in all CD22KO replicates, while Nnt showed undetectable expression in all WT replicates. N.D. = not detectable. Statistical analyses are two-way ANOVA A–D). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. All other comparisons not significant ($P \geq 0.05$).

PD-L2, and Arginase (Fig. 2D). Overall, CD22KO BMDMs showed largely similar expression of M1-like polarization markers and M2-like polarization markers, measured by both protein and transcript expression, after 48 h of polarization to either polarization state.

For an initial view of function, we next compared common cytokines secreted by M2-like BMDMs in 48 h-polarized WT and CD22KO BMDMs. As was observed with polarization marker expression, CD22KO 48 h-polarized M2-like BMDMs showed no difference in production of IL-10, IL-12p70, GM-CSF, and VEGF (Fig. 2E). Taken together, these surface marker and cytokine results indicate that loss of CD22 does not impact the ability of BMDMs to polarize to an M1-like or an M2-like phenotype.

CD22KO M2-like BMDMs exhibit transcriptional changes

Our previous findings demonstrated that CD22 expression in BMDMs is high after 48 h of IL-4 stimulation to induce an M2-like phenotype (Kukan et al. 2025). To determine the transcriptional impact of CD22 ablation on M2-like macrophages, we utilized RNA sequencing (RNAseq). First, with our overall interest in the recognition and synthesis of sialylated glycans, we specifically sought to compare expression levels of the Siglec and sialyltransferase families. We found that M2-like CD22KO BMDMs had increased Siglec-E and decreased Siglec-3, although these changes were relatively modest (Fig. 3A). Expression of Siglec-1 (sialoadhesin/CD169) was strong regardless of the presence or absence of CD22 (Siglec-2). Siglec-4 and Siglec-F were also expressed, but again no difference was observed. Similarly, we observed modest decreases in expression of St3gal5 and St8sia4 sialyltransferases but no change among other expressed sialyltransferases such as St3gal1 and St6gal1 (Fig. 3B).

Corroborating our flow cytometry findings (Fig. 2), M2-like CD22KO BMDMs showed only modest differences in polarization marker transcripts compared to WT, such as CD86 and CD206, exhibiting no cohesive pattern in differential expression (Fig. 3C and D). CD22KO BMDMs showed small decreases in expression of iNOS, a canonical M1-like marker, and a concomitant increase in arginase, an M2-like marker, while CD22KO BMDMs showed decreased expression of Ym1, another M2-like marker. Thus, while slight changes are observed on a gene-by-gene basis, no discernable shift in polarization state is seen in 48 h-polarized M2-like CD22KO BMDMs as compared to WT BMDMs (Figs. 2 and 3A–D).

Moving away from pre-selected genes of interest, we also compared the complete transcriptomes of CD22KO and WT BMDMs. We curated a list of 22 differentially expressed genes (DEGs) with appreciable expression in our WT or CD22KO 48-h polarized M2-like BMDMs from a broader list of 74 overall identified DEGs, with some increased in CD22KO and others decreased (Fig. 3E). These data suggest that loss of CD22 leads to broader changes in cellular function that are not readily apparent in simplistic polarization data covering a few key markers.

Loss of CD22 drives broad transcriptional changes in M2-like BMDMs

To further interrogate the broader changes in the set of differential genes identified within the RNAseq data, we utilized Gene Set Enrichment Analysis (GSEA) to identify pathway differences driven by loss of CD22 in BMDMs. This unbiased analysis, which takes into account compounding small changes in expression of a defined set of genes, revealed several notable changes. We found that CD22KO BMDMs were enriched for pathways such as “Asthma” and “Allograft Rejection,” characteristic of a pro-inflammatory transcriptome shift (Fig. 4A and B). Genes identified as significant contributors of enrichment included *Tnf* and *CD40* for “Asthma” and many antigen presentation glycoproteins, such as the MHC-II molecules *H2-Dma*, *H2-DMb1*, and *H2-Dmb2*, for “Allograft Rejection.”

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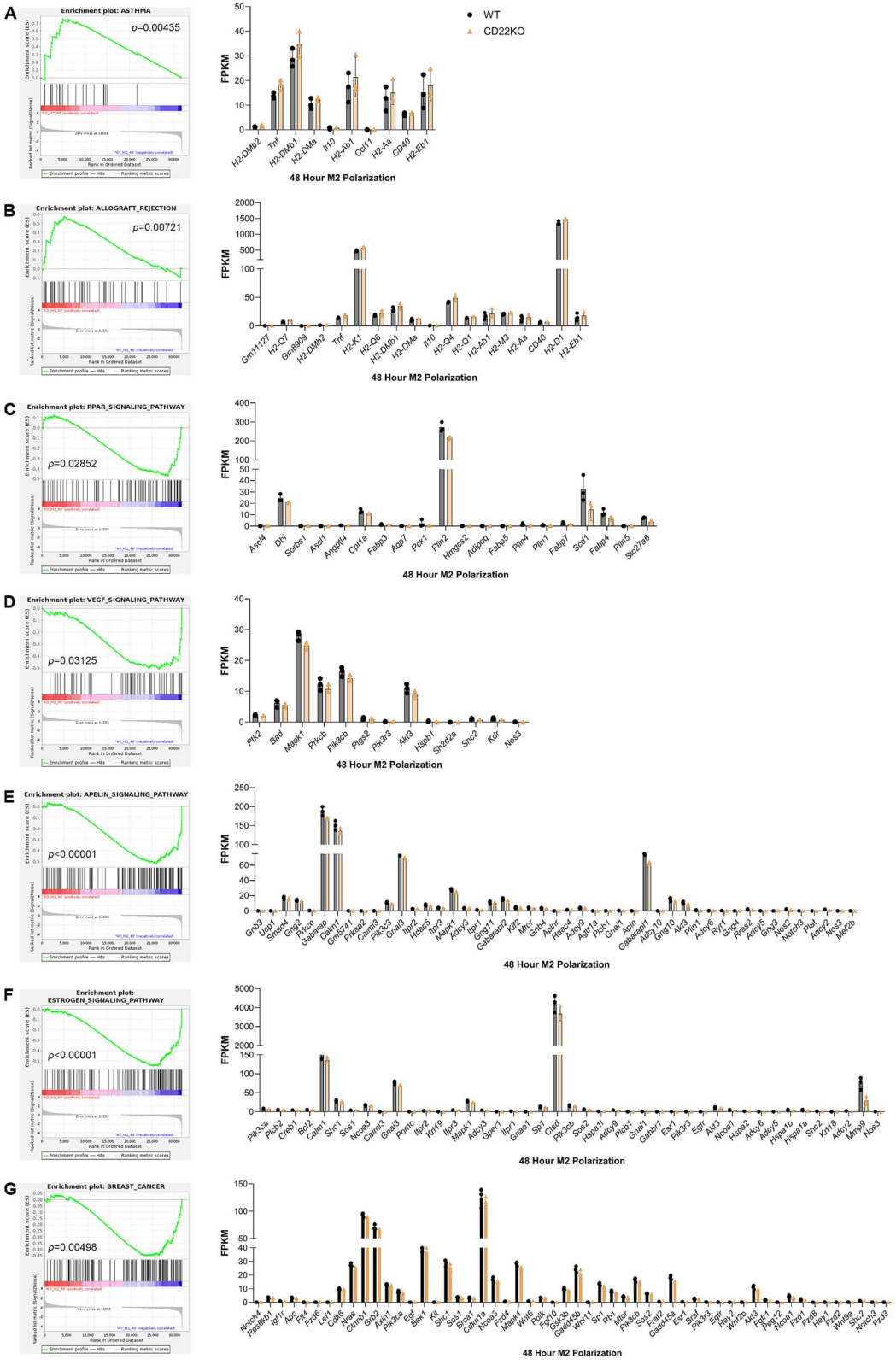


Fig. 4. Loss of CD22 drives broad transcriptional changes in M2-like BMDMs. GSEA plots and FPKM values for associated core enriched genes for "asthma" A), "allograft rejection" B), "PPAR Signaling Pathway" C), "VEGF Signaling Pathway" D), "Apelin Signaling Pathway" E), "Estrogen Signaling Pathway" F), and "breast cancer" G) from 48-h M2-polarized WT and CD22KO BMDMs ($n = 3$).

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Augmenting this apparent shift towards inflammatory responses, M2-like CD22KO BMDMs were significantly less enriched for several anti-inflammatory pathways. Two of these pathways were the "PPAR Signaling Pathway" and "VEGF Signaling Pathway" (Fig. 4C and D), both of which are particularly interesting because they are known to be critical for M2-like programming of macrophage function (Odegaard et al. 2007; Wheeler et al. 2018). For example, WT M2-like BMDMs were enriched for genes such as *Fabp4* encoding the adipocyte fatty acid-binding protein, which has been shown to promote breast cancer survival when upregulated in tumor-associated macrophages (TAMs) (Hao et al. 2018). Similarly, other anti-inflammatory pathways showed significant enrichment in the WT cohort (i.e. were reduced in the CD22KO BMDMs), including the "Apelin Signaling," "Estrogen Signaling," and "Breast Cancer" pathways (Fig. 4E–G).

Collectively, and consistent with the notion that CD22 and other Siglec family members are generally important regulatory receptors which dampen inflammation (Lübbers et al. 2018), the changes in the identified pathways based on unbiased GSEA point to significant decreases in anti-inflammatory pathways and concomitant enrichment in pro-inflammatory pathways upon ablation of CD22 in M2-like-skewed BMDMs. This suggests that abrogation of CD22 in M2-like BMDMs results in an overall pro-inflammatory transcriptome shift. However, it is also important to note that these changes are observed at the transcript level; whether these transcriptional changes lead to commensurate changes in protein expression warrants future study.

Loss of CD22 impairs M2-associated macrophage functions

We previously identified CD22 as an important mediator of endocytic processes in macrophages, assisting in the endocytosis and degradation of sialylated glycoproteins as well as efferocytosis of apoptotic bodies (Kukan et al. 2025), which suggested that CD22 may participate in the immunologically-silent housekeeping cleaning of localized debris. To assess these processes in a CD22KO background, we polarized WT and CD22KO BMDMs to an M2-like phenotype for 48 h as before and performed similar functional analyses.

First, to test the endocytic behavior of the cells, BMDMs were incubated with labeled native fetuin or fetuin treated with α2,3,6,8 neuraminidase to remove all sialic acids. We found that the removal of sialylation significantly reduced endocytosis in both WT and CD22KO BMDMs (Fig. 5A, Fig. S3) by about 50%. This is consistent with our findings demonstrating that both WT and CD22KO cells express several Siglec family members to roughly equal extents (Fig. 3A) and that the neuraminidase removes all forms and linkages of sialylation. We also found that ablation of CD22 specifically reduced endocytosis compared to WT BMDMs (Fig. 5A) by approximately 50%.

In B cells, CD22 is localized within nanodomains, often engaging in cis interactions with α2,6-sialylated ligands on the surface of the cell (Gasparrini et al. 2016). These cis interactions can "mask" CD22 from trans ligands in resting B cells (Razi and Varki 1998), although cis interactions do not appear to inhibit CD22's ability to redistribute on the surface on the B cell in response to trans interactions (Collins et al. 2004) nor its constitutive endocytosis in B cells (Zhang and Varki 2004). Thus, we investigated the extent to which cis interactions on the surface of macrophages affect endocytosis facilitated by CD22 by comparing fetuin endocytosis in 48 h-polarized M2 WT BMDMs and BMDMs from ST6Gal1 knockout mice lacking α2,6-sialyation (Fig. 5B). Interestingly, ST6KO BMDMs exhibited slightly decreased endocytosis of sialylated fetuin, while WT and ST6KO BMDMs showed no difference in endocytosis of desialylated fetuin, suggesting either CD22 is minimally masked by cis ligands or CD22-mediated endocytosis in trans is unaffected by cis interactions on the surface of BMDMs (Fig. 5C, Fig. S3).

Next, we assessed the extent to which CD22 ablation impacts efferocytosis, a process whereby macrophages endocytose apoptotic debris from environments to mitigate further inflammation. We harvested recruited peritoneal cells which were primarily neutrophils from mice and induced apoptosis overnight through staurosporine treatment. After labeling the apoptotic "bait" cells, we incubated them with M2-like WT or CD22KO BMDMs to assess uptake via flow cytometry. Consistent with the 50% reduction in endocytosis (Fig. 5A), CD22KO BMDMs had a significant 50% reduction in efferocytosis as compared to their WT CD22$^+$ counterparts (Fig. 5D, Fig. S3).

Although we do not claim that all endocytosis and efferocytosis that occurs in BMDMs of any polarization state is solely a result of Siglec-glycan interactions, we also probed the binding specificity of CD22 to validate the importance of sialic acids. Indeed, our previous findings utilizing ST6Gal1 knockout bait cells suggested that CD22 had some capacity to bind glycans even without the presence of α2,6-linked sialic acids (Kukan et al. 2025). Using a glycan binding array at the National Center for Functional Glycomics, we found that the top ligands for murine CD22 ectodomain were sialylated species as expected, although binding was also seen with non-sialylated glycans (Fig. 5E). Interestingly, the second-best ligand on the array carried α2,3-linked sialic acids, whereas other top ligands had terminal galactose, GlcNAc, and fucose residues without sialic acid present, illustrating that CD22 has some affinity for glycans other than α2,6-sialylated species. Most other glycoforms failed to significantly interact with CD22 on the array (Fig. S4 and Table S1). Although somewhat surprising, these data align with prior NCFG array data for human CD22 binding (https://www.functionalglycomics.org/glycan-array; primary screen ID: 1792) showing both a top hit for an α2,6-sialylated species with an underlying sulfated GlcNAc residue, as well as several instances of CD22 binding to non-sialylated glycans within the top 10 hits. Overall, these data confirm that while CD22 prefers sialylated ligands, binding can occur with non-sialylated glycans, and thus CD22 can contribute to endo/efferocytotic function, albeit to a lesser degree, even without sialic acids present.

Finally, having established reduced sialic acid-dependent and independent endo/efferocytosis in CD22KO BMDMs, we assessed the extent to which CD22 ablation impacts degradation of internalized proteins using a DQ-Ova assay and flow cytometry (Lewis and Cobb 2010). We found that M2-like CD22KO BMDMs had significantly reduced degradation signal, whether from reduced endocytosis and/or reduced protein cleavage (Fig. 5F, Fig. S3). We also confirmed that the signal was due to acid-dependent lysosomal proteolytic cleavage using Bafilomycin A1 (BafA) to block lysosomal acidification. As expected, no signal was seen in BafA-treated cells. Taken together, our findings indicate that CD22 ablation

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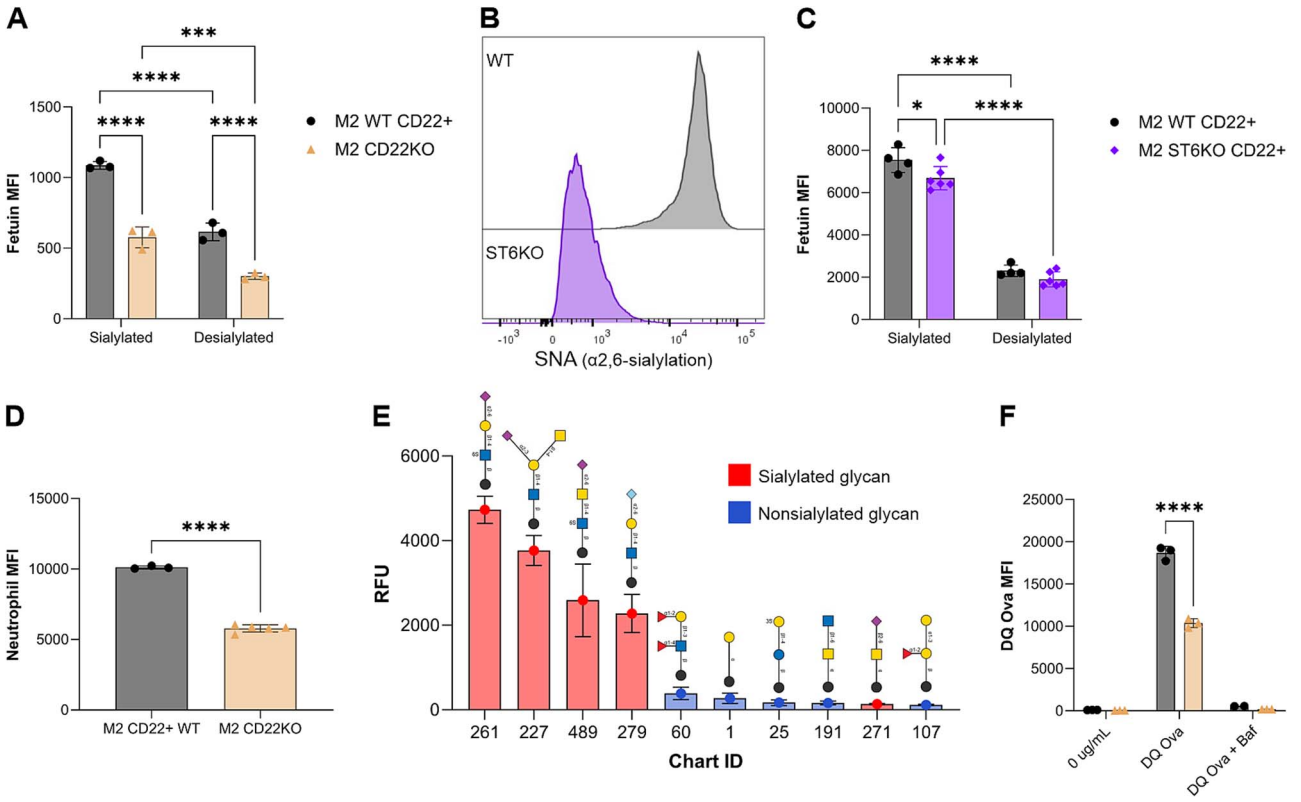


Fig. 5. Loss of CD22 impairs endocytosis, protein degradation, and efferocytosis in M2-like BMDMs. A) Flow cytometry data showing MFI of sialylated or desialylated fetuin endocytosed by CD22$^+$ WT or CD22KO BMDMs polarized to M2-like for 48 h ($n = 3$). B) Representative flow histograms of WT and ST6KO BMDMs polarized to M2-like for 48 h stained with SNA. C) Flow cytometry data showing MFI of sialylated or desialylated fetuin endocytosed by CD22$^+$ WT or ST6KO BMDMs polarized to M2-like for 48 h (N = ≥ 4). D) Flow cytometry data of MFI of neutrophils efferocytosed by 48 h-polarized M2-like WT CD22$^+$ or CD22KO BMDMs ($n \geq 3$). E) Binding data (relative fluorescence units (RFU)) of top ten glycan hits for recombinant murine CD22 ectodomain-fc reagent and associated glycan structures ($n = 3$). F) Flow cytometry data showing MFI of DQ ova degraded by 48-h M2-polarized WT CD22$^+$ or CD22KO BMDMs with or without BafA treatment ($n = 3$). Statistical analyses are two-way ANOVA A, C, F) or Student's t-test D). $^{***}P < 0.001$, $^{****}P < 0.0001$. All other comparisons not significant ($P \geq 0.05$).

in BMDMs results in functional differences characterized by reduced endocytic, degradative, and efferocytotic capacities.

Discussion

Environmental sialylation can drive robust changes in cellular and tissue phenotype. Highly α2,6-sialylated tissue environments tend to be more immune-tolerant, with their resident macrophage populations showing a higher threshold for activation (Martin et al. 2021; Liu et al. 2023). Sialic acids, in particular the α2,6 linkage, are therefore viewed as "anti-inflammatory switches" or "self-associated molecular patterns," representing a glycobiological axis through which tissues and cells can promote tolerogenic environments and responses (Varki 2011; Paul and Padler-Karavani 2018).

Siglecs, a family of receptors whose primary ligands are glycans with terminal sialic acid residues, are critical modulators of phenotype and function across a variety of immune cells, often mediating immunomodulatory or anti-inflammatory functions. Siglec-2 (CD22) is widely recognized as a key negative regulator of B cells, preventing autoreactive and overreactive B cell responses through its recognition of α2,6-linked sialic acids with concurrent B cell receptor (BCR) ligation. Recently, CD22 has been investigated for its role in macrophages, a cell type not previously thought to express CD22. Several reports have demonstrated that CD22 inhibits phagocytosis in microglia (Pluvinage et al. 2019; Aires et al. 2021), and our group has recently demonstrated that CD22 promotes endocytosis and efferocytosis of sialylated debris in BMDMs and is a potent characteristic of tissue-resident macrophages in highly sialylated and immune-privileged tissues such as the lung and liver (Kukan et al. 2025).

In the present study, we demonstrate that genetic loss of CD22 has broad impacts on macrophage phenotype and function. When in a liver environment lacking α2,6-linked sialylation on hepatocytes (HcKO mouse), a pro-inflammatory shift among tissue-resident macrophages precedes changes in the systemic (splenic) T cell compartment characterized by a loss of naïve T cells and a gain in antigen-experienced T cells (Oswald et al. 2020a; Oswald et al. 2020b). Here, we found that T cells isolated from young versus old mice reflected this shift in effector class (Fig. 1A and B). More importantly, the ablation of α2,6-sialylation in the liver and the resulting increase in antigen-experienced T cells and eventual fatty liver disease is phenocopied by the reciprocal ablation of its receptor CD22 (Fig. 1E), strongly suggesting that the sialic acid-CD22 axis among tissue-resident macrophages is a potent and critically important regulatory pathway that drives homeostasis.

Interestingly, macrophages derived from CD22KO animals (CD22KO BMDMs) showed similar expression of M1-like and M2-like markers as compared to WT BMDMs upon

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polarization in vitro (Fig. 2A–D). M2-like CD22KO BMDMs also exhibited no difference in production of M2-associated cytokines such as IL-10 and VEGF (Fig. 2E). RNAseq analysis confirmed these results. The transcriptional expression of several Siglecs and sialyltransferases was modestly altered, but no clear differences in an array of M1-like and M2-like polarization markers were observed (Fig. 3). Collectively, these data indicate that CD22KO does not drive a shift in canonical polarization state as defined by traditional M1-like and M2-like markers.

Although the canonical polarization markers remained similar and no clear shift in pro- or anti-inflammatory phenotype was seen by these biased analyses, unbiased GSEA from RNAseq data revealed important changes in the CD22KO BMDMs (Fig. 4). Ablation of CD22 showed significant enrichment of several KEGG-defined pro-inflammatory pathways, including "Asthma" and "Allograft Rejection," accompanied with decreased enrichment of key anti-inflammatory pathways such as "VEGF Signaling" and "PPAR Signaling." Decreased expression of *Mmp9* (highlighted in the Estrogen Signaling Pathway gene set) is indicative of a reduced M2-like functionality, as matrix metalloproteinase 9 (MMP9) is a key mediator of the wound healing process and has been directly implicated in tumor metastasis (Tian et al. 2022). These findings suggest that loss of CD22 results in broad transcriptional changes in M2-like macrophages toward a more pro-inflammatory phenotype, similar to what was observed in GSEAs of our HcKO model (Oswald et al. 2020a).

Consistent with the transcriptomic differences, ablation of CD22 also resulted in impaired M2-associated macrophage function. CD22KO M2-polarized BMDMs phenocopied WT CD22- M2-polarized BMDMs (Kukan et al. 2025) with reduced endocytic capacity, reduced protein degradation, and reduced efferocytotic ability (Fig. 5). Given that CD22 is a known endocytic receptor in B cells, these data underscore the importance of CD22 as a key player in macrophage endocytosis and efferocytosis, particularly in M2-like populations, where it may also mark M2 macrophages with enhanced overall endocytic behavior. It is important to note that these observations are specific to murine CD22 and cannot be extrapolated to human macrophages due to the different ligand preferences of murine and human CD22 (Blixt et al. 2003; Crocker et al. 2007). CD22 expression in human macrophages remains largely unexplored, although expression has been demonstrated in human induced pluripotent stem cell-derived macrophages (Aires et al. 2021). Thus, exploration of CD22 expression in human M2-like macrophages represents an salient avenue for future investigation, especially given the upregulation of other inhibitory Siglecs in human M2-like macrophage populations, such as Siglec-10 (Yang et al. 2025). Interestingly, removing cis ligands on the surface of BMDMs through our ST6KO model did not appreciably impact endocytosis (Fig. 5C). This result may be indicative that CD22 exhibits differential cell surface localization and distribution in macrophages than in B cells, warranting further studies of CD22's cell surface dynamics in macrophages.

Our findings support a model in which CD22 both identifies a population of M2-like macrophages skewed to promote clearance of self-debris characterized predominantly by the presence of sialylation and directly participates in the uptake of that debris in a fashion that fails to promote inflammation. The presence of CD22 in sialylation-rich environments, such as the liver, helps to maintain a lack of antigen presentation to and activation of T cells, yet loss of either the ligand (sialylated glycans) or their receptor (CD22) shifts the systems towards promoting antigen presentation and T cell activation. Indeed, CD22 expression was mostly found in macrophages low in costimulatory molecules, such as CD86, expression, although ablation of CD22 does not appear to directly impact macrophage polarization as judged by the usual markers. This suggests that the increased number of pro-inflammatory-skewed macrophages in the HcKO mouse model livers as well as the development of a similar fatty liver phenotype in the CD22KO mouse model is due to proliferation and recruitment of M1-like macrophages rather than a shift in polarization of existing resident macrophages.

Finally, it is important to recognize that CD22 is a potent signal transducer. The cytoplasmic tail contains four ITIM/ITIM-like motifs that upon phosphorylation in B cells assist in recruiting phosphatases, such as SHP-1, to proximal signaling complexes (Tedder et al. 2005). The extent to which CD22 transmits a signal within macrophages and what this does to the function of the cells under a variety of conditions is completely unknown but is ultimately critical for understanding the mechanisms underlying the functional impact of CD22 in macrophages.

Materials and methods

Mouse strains

All protocols for these studies were approved by Case Western Reserve University (CWRU) IACUC. All wild-type (WT) mice used in these studies were B6.129-*St6gal1*^{tm2Jxm}/J mice from The Jackson Laboratory (Bar Harbor, ME) (Strain #006091) (Hennet et al. 1998) housed in the CWRU Animal Research Facility (ARC). All HcKO mice used in these studies were housed in the CWRU ARC and generated as previously described (Oswald et al. 2020a). All CD22 knockout mice (CD22KO) used in these studies were C57BL/6-Cd22^{tm1Lam}/J mice from The Jackson Laboratory (Strain #006940) housed in the CWRU ARC (Nitschke et al. 1997). All ST6Gal1 knockout mice (ST6KO) used in this study were derived from crosses of the WT mice described above with B6.C-Tg(CMV-cre)1Cgn/J mice from The Jackson Laboratory (Strain 006054) (Schwenk et al. 1995). Mice were age and sex-matched for all experiments.

Data and statistical analyses

All data and statistical analyses were plotted and performed using GraphPad Prism 10.2.3 (GraphPad Software). Statistical analyses include Student's t-test and two-way ANOVA where appropriate. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$.

Bone marrow-derived macrophage (BMDM) media

All BMDM experiments were performed in BMDM media as follows: Advanced RPMI-1640 (Gibco, Thermo Fisher Scientific) supplemented with 10% fetal bovine serum (FBS) (Biowest), 1% Penicillin/Streptomycin (Thermo Fisher Scientific), and 1% L-glutamine (Thermo Fisher Scientific) or RPMI-1640 (HyClone) supplemented with 10% FBS, 1% Penicillin/Streptomycin, 1% L-glutamine, 1% MEM non-essential

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amino acids (Thermo Fisher Scientific), and 1% sodium pyruvate (Thermo Fisher Scientific).

BMDM culture

Mice were sacrificed via CO_2 inhalation according to an IACUC protocol approved by CWRU. Nested microcentrifuge tubes were prepared with a hole poked in the inner tube. Femurs were spun in the inner tube for 10 min at 2000xg to spin out marrow. Bone marrow pellets were resuspended in BMDM media, counted, and plated in 10 cm petri dishes at 5 million cells per plate. Cells were plated in 7.5 mL BMDM media and 2.5 mL L929-conditioned media. Cells were incubated at 37 °C and 5% CO_2. Media was supplemented on Day 3 via addition of 2 mL BMDM media and 1 mL L929-conditioned media per plate. All media was removed from plates on Day 6 and replaced with 7.5 mL fresh BMDM media and 2.5 mL fresh L929-conditioned media. Cells were washed with 1X PBS (Gibco, Thermo Fisher Scientific), scraped, and utilized for experiments on Day 7.

BMDM polarization

Cells were seeded following harvest on Day 7 as described above. All conditions were polarized in media containing 25% L929-conditioned media. Cells were polarized to M1-like utilizing 5 ng/mL LPS (Invivogen) and 50 ng/mL IFNγ (Peprotech, Thermo Fisher Scientific). Cells were polarized to M2-like utilizing 20 ng/mL IL-4 (R&D Systems). M0-like cells received no additional stimulus. Polarizations were carried out for the indicated amount of time at 37 °C and 5% CO_2.

Tissue histology

Mice were sacrificed by CO_2 inhalation. For H&E staining, tissues were harvested and fixed in 10% formalin (VWR) for 24 h. Tissues were then embedded and mounted on slides by AML Laboratories (Jacksonville, FL). For Oil Red O staining, tissues were snap frozen in liquid nitrogen; staining and mounting were performed by the Tissue Resources Core at University Hospitals (Cleveland, OH). All histology images were acquired via the TCS SP5 confocal microscope (Leica).

T cell activation assay and IFNγ ELISA

96 well plates were coated with anti-CD3 (eBioscience) at the indicated concentrations overnight at 4 °C. To obtain $CD4^+$ T cells, mouse spleens were harvested, mashed through a 100 μm filter, incubated with CD4 magnetic beads (Miltenyi Biotec), and passed over an LS column (Miltenyi Biotec). 50,000 $CD4^+$ T cells were added to each well and incubated for 72 h at 37 °C. Culture supernatants were analyzed via IFNγ ELISA kit (BioLegend) per manufacturer's instructions. In brief, capture antibody was coated overnight at 4 °C, wells were blocked for one hour at room temperature (RT), culture supernatants were incubated for two hours at RT, detection antibody was added for one hour at RT, Europium-conjugated streptavidin (1:1000 in Europium dilution buffer) (Revvity) was added for one hour at RT, and enhancement solution (Revvity) was added for five minutes before reading on a VICTOR Nivo Plate Reader (Revvity). Wells were washed three times with PBS-T between steps.

Flow cytometry

Cells were harvested and blocked using 1:1000 Fc block or 1X Carbo-Free™ Blocking Solution (Vector Labs) in PBS for 15 min on ice. Cells were stained with indicated antibodies or lectins for 30 min on ice. Cells were washed with PBS, and flow cytometry was run on an Attune NxT, BD LSRFortessa, or FACSAria-SORP cytometer (Case Cytometry & Imaging Microscopy Core). Data were analyzed using FlowJo v10.8.0 (FlowJo LLC) and are displayed as percent positive or median fluorescent intensity (MFI).

Reagents used:

BV711 anti-mouse F4/80 – Clone T45–2342 (BD Biosciences).

Biotin anti-mouse Ly6C – Clone HK1.4 (BioLegend).

APC-efluor 780 streptavidin (eBioscience).

APC anti-mouse CD22 – Clone OX-97 (BioLegend).

BV421 anti-mouse CD22 – Clone Cy34.1 (BD Biosciences).

FITC anti-mouse F4/80 – Clone BM8 (BioLegend).

BUV395 anti-mouse F4/80 – Clone T45–2342 (BD Biosciences).

PE-Cy7 anti-mouse CD206 – Clone C068C2 (BioLegend).

APC anti-mouse CD301 – Clone LOM-14 (BioLegend).

PE anti-mouse CD86 – Clone GL1 (BD Biosciences).

SNA-FITC (Vector Labs).

T cell Co-culture

To obtain liver macrophages, livers were perfused and digested with 1X collagenase D (Roche, MilliporeSigma). Cells were separated via Percoll gradient (Sigma-Aldrich, MilliporeSigma) and sorted via CD11b magnetic beads (Miltenyi Biotec) and LS columns (Miltenyi Biotec). $CD11b^+$ cells were further sorted (staining protocol described above) using a FACSAria-SORP cytometer to collect $F4/80^+Ly6C^-$ cells (Kupffer cells). $CD4^+$ T cells were obtained from the spleen of an OT-II mouse as described above. For co-culture, 50,000 $CD4^+$ T cells were plated with 100,000 Kupffer cells with or without Ova protein (50 μg/mL) and incubated for 72 h at 37 °C. Culture supernatants were analyzed by Luminex, described below.

Luminex cytokine analysis

Luminex cytokine analysis was performed by Eve Technologies (Calgary, Alberta, Canada) on culture supernatants. Cytokines were analyzed using the 32-plex analysis.

RT-qPCR

RNA was isolated from cells using the RNEasy Mini RNA Isolation Kit (Qiagen) per manufacturer's instructions. RNA was converted to cDNA using RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific). cDNA was used for qPCR with Taqman Fast Advanced Master Mix (Thermo Fisher Scientific). ΔCT was plotted relative to beta-2 microglobulin (β2M).

Primers used (Taqman):

β2M: Mm00437762_m1.

CD22 (exon 9–11): Mm00515438_m1.

CD86: Mm00444540_m1.

PD-L2: Mm00451734_m1.

CD206: Mm01329359_m1.

CD301a: Mm00546125_g1.

Arginase: Mm00475988_m1.

iNOS: Mm00440502_m1.

RNA sequencing

Poly-A RNA sequencing, differential gene analysis, GSEA analysis, and GSEA plot generation were performed by LC Sciences. Differentially expressed gene (DEG) heatmap (Fig. 3E)

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was generated by normalizing CD22KO replicates to WT replicates and color coding based on $\log_{10}$ of this normalization and at a range of minimum (low; blue) to maximum (high; red) within the samples shown. The curated list of DEGs was generated by including hits with any replicate having greater than 1 FPKM expression from a list of 74 identified DEGs provided by LC Sciences analysis. Three biological replicates of WT and three biological replicates of CD22KO BMDMs were utilized for these analyses. All BMDM samples were harvested, differentiated, and polarized to M2-like for 48 h using IL-4 as described above. All GSEAs were generated using KEGG pathway analysis.

Fetuin endocytosis assay

Fetuin (Sigma-Aldrich, MilliporeSigma) desialylation proceeded overnight at 37 °C using α2,3,6,8 neuraminidase (New England BioLab) per manufacturer's instructions. 500 μg of each fetuin (native (sialylated) and desialylated) was labeled with AF488 per manufacturer instructions using the Alexa Fluor™ 488 Microscale Protein Labeling Kit (Thermo Fisher Scientific). 6 μg of the respective fetuin was added to 48 h-polarized M2-like BMDMs and incubated for 1 h. Cells were harvested, washed, and prepared for flow cytometry.

DQ-ova assay

BMDMs were polarized to M2-like phenotype for 48 h as described above. Cells were washed twice with PBS and incubated with 6.25 μg/mL DQ-Ova solution (Thermo Fisher Scientific) with or without 200 nM Bafilomycin A1 (Sigma-Aldrich, MilliporeSigma) for 1 h at 37 °C. Following incubation, cells were washed with PBS, collected, and prepared for flow cytometry as described above.

Efferocytosis assay

BMDMs were polarized to M2-like phenotype for 48 h as described above. To acquire peritoneal cells, mice were injected intraperitoneally with 1 mL 4% thioglycollate broth (Sigma-Aldrich, MilliporeSigma) 48 h before assay. Mice were sacrificed 24 h after injection. Recruited peritoneal cells were collected via peritoneal lavage with 5 mL sterile PBS. Apoptosis of cells was induced overnight utilizing staurosporine per manufacturer instructions using the Efferocytosis Assay Kit (Cayman Chemical). Bait cells were harvested and CFSE stained per manufacturer instructions. BMDMs were concurrently harvested and stained with F4/80 and CD22 as described above. Following staining, cells were washed. Cells were plated in 1:1 ratios of bait cells:macrophages and incubated at 37 °C for indicated times in BMDM media. Following incubation, cells were harvested, washed, and prepared for flow cytometry.

Lectin ELISA

Lectin ELISAs were performed on plasma samples as previously published (Oswald et al. 2019). Briefly, plasma was extracted from mice via cardiac puncture, and 1:10 sodium citrate was added to prevent clotting. Blood was centrifuged at 2500 × g for 15 min to remove all cells, and the supernatant (i.e. the plasma) was collected. Plasma was coated overnight at 4 °C on 96 well ELISA microplates (Greiner Bio-One). Wells were washed three times with PBS-T and blocked with 1X Carbo-Free™ Blocking Solution (Vector Labs) for one hour at RT, then washed three times with PBS-T. Lectin stain was added to the plate (2 mg/mL), and staining proceeded overnight at 4 °C. Wells were washed three times with PBS-T, and Europium-conjugated streptavidin (Revvity) (1:1000 in Europium dilution buffer) was added to incubate for one hour at RT. Wells were washed three times with PBS-T, and enhancement solution (Revvity) was added for five minutes before reading using a VICTOR Nivo Plate Reader (Revvity).

Lectins used: ConA-Biotin, ECL-Biotin, LCA-Biotin, MAL-I Biotin, PHA-E-Biotin, PHA-L-Biotin, and SNA-Biotin (Vector Labs).

Glycan Array

Glycan array data was performed at the National Center for Functional Glycomics (NCFG) using 50 μg recombinant murine CD22 ectodomain-human Fc reagent (Sino Biological) and the Consortium for Functional Glycomics glycan array. Images of glycan structures were prepared using the GlycoGraph tool (Mehta and Cummings 2020).

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Author contributions

Emily Kukan (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Funding acquisition [supporting], Investigation [equal], Methodology [equal], Project administration [equal], Visualization [lead], Writing—original draft [equal], Writing—review & editing [equal]), Gabrielle Fabiano (Formal analysis [equal], Investigation [equal]), Julie Zhou (Conceptualization [supporting], Formal analysis [supporting], Investigation [supporting]), Ava Richard (Investigation [supporting]), Kevin Telfer (Conceptualization [supporting], Formal analysis [supporting], Investigation [supporting]), Brian A Cobb (Conceptualization [equal], Data curation [equal], Formal analysis [supporting], Funding acquisition [lead], Investigation [supporting], Methodology [equal], Project administration [lead], Visualization [supporting], Writing—original draft [equal], Writing—review & editing [equal]).

Supplementary material

Supplementary material is available at *Glycobiology* online.

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Conflicts of interest

None declared.

Data availability

All original data are embedded in the Figures of this manuscript or provided within the Supplementary Materials with the exception of the

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RNAseq data, which is available through the Gene Expression Omnibus at the NCBI under the accession number GSE310152.

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